

DSA 8401: Applied Machine Learning in Data Science
Capstone Project Status Report
Mortality Prediction in ICU Patients with Acute Kidney Injury (AKI)

Date: 6th October 2025

1. Project Overview

This capstone project focuses on developing a **machine learning model to predict in-hospital mortality (IHM)** among ICU patients diagnosed with **Acute Kidney Injury (AKI)** using data derived from the **MIMIC-III database**. The project implements a complete data science pipeline—from preprocessing to model evaluation—designed to optimize predictive performance while maintaining interpretability and clinical relevance.

Our predictive model aims to assist clinicians by leveraging routinely collected ICU data, including demographic variables, vital signs, and laboratory results such as **heart rate, blood pressure, temperature, WBC count, BUN, sodium, potassium, bilirubin, FiO₂, PaO₂**, and the **Glasgow Coma Scale (GCS)**.

2. Work Completed

2.1 Data Understanding and Cleaning

- i. Imported and explored MIMIC-III-based AKI data, inspecting data types, distributions, and missingness.
- ii. Performed **data cleaning** to handle missing values using **median and mode imputation** strategies.
- iii. Detected and treated outliers using **interquartile range (IQR)** capping to avoid biasing model learning.
- iv. Normalized numerical variables (z-score scaling) to improve model convergence and comparability.
- v. Encoded categorical variables (e.g., Sex) using **one-hot encoding** for compatibility with ML algorithms.

2.2 Exploratory Data Analysis (EDA)

- i. Conducted detailed statistical and visual exploration to uncover relationships between predictors and IHM.
- ii. Found strong mortality associations with **high BUN, low GCS, elevated bilirubin, and low oxygen ratios (PaO₂/FiO₂)**.
- iii. Generated correlation heatmaps, boxplots, and pairplots to visualize variable interactions and confirm clinical relevance.

2.3 Feature Engineering and Class Balancing

- i. Addressed **class imbalance** (mortality cases < 20%) using **Synthetic Minority Oversampling Technique (SMOTE)**, improving the model's ability to learn minority class patterns.
- ii. Selected clinically relevant and statistically significant features to prevent overfitting.
- iii. Split the dataset into **80% training** and **20% testing** sets using **stratified sampling** to maintain class proportions.

2.4 Model Development

- i. Implemented and compared multiple models:
 - o **Logistic Regression** – baseline interpretable model.

- **Random Forest Classifier** – strong performance with interpretability.
- **XGBoost Classifier** – optimized gradient boosting for highest predictive power.
- ii. Applied **GridSearchCV** for hyperparameter optimization (e.g., tree depth, learning rate, estimators).
- iii. Evaluated models using **ROC-AUC, F1-score, Precision, and Recall** metrics.

2.5 Results and Evaluation

- i. **XGBoost** achieved the best results with a **ROC-AUC ≈ 0.89** and **F1-score ≈ 0.81** on validation data.
- ii. **Feature importance** analysis showed that **GCS, BUN, bilirubin, and FiO₂** were the top predictors of mortality.
- iii. Generated ROC and precision-recall curves to visualize performance across thresholds.
- iv. Identified an **optimal probability threshold** that maximized the F1-score, as required by the capstone evaluation.

3. Next Steps

1. **Finalize Threshold Optimization** – Confirm the threshold that yields the best F1-score and apply it consistently in the final model submission.
2. **Finalize and Test the Notebook** – Ensure the Colab notebook runs end-to-end with clear documentation and reproducible results.
3. **Prepare the Final Submission Materials** – The notebook link will be included in the submission PDF as per the course guidelines.
4. **Refine the Draft Presentation (PPT)** – A draft presentation has already been created, summarizing data exploration, model comparison, results, and each team member's contributions. The next step is to refine it to fit the 10-minute format and add visuals for interpretability.

5. Challenges and Solutions

Challenge	Description	Solution Implemented
Class Imbalance	Mortality cases were underrepresented.	Applied SMOTE oversampling to improve sensitivity.
Missing Data	Several lab variables had incomplete entries.	Used median/mode imputation based on data type.
Overfitting	Complex models risked memorizing training data.	Used cross-validation and tuned regularization parameters.
Threshold Selection	Need to maximize F1-measure for evaluation.	Conducted precision-recall analysis to find optimal cutoff.

5. Conclusion

To date, the project team has successfully completed data preparation, exploratory analysis, feature engineering, model development, and internal validation. The **XGBoost classifier** currently offers the most balanced and high-performing solution. The next phase will focus on finalizing the probability threshold and preparing the formal presentation and submission package. (Notebook: <https://drive.google.com/file/d/1pyeVFrX4ISuciS4riEwRkuN1Mta82LTW/view?usp=sharing>)